Colon adenocarcinoma

2nd leading cause of cancer death · ~106,590 new cases (2024) · 5-yr survival — stage I **90%**, II **70–85%**, III **50–73%**, IV **12–14%**

Stanford Oncology Handbook 2025, Colon Cancer (Johnson T, Miller D) · NCCN · items marked † are outside the handbook chapter

WORKUP

- Colonoscopy with biopsy + pathology review
- **MMR** / **MSI testing in ALL newly diagnosed** colon cancer
- **KRAS, NRAS, BRAF** mutation + **HER2** amplification by NGS panel
- CT C/A/P · MRI liver if potentially resectable liver mets · PET/CT in selected resectable M1
- Labs: CBC, CMP, **CEA**
- Genetics referral: all diagnosed **<50 y**, or somatic finding with germline implications
- BRAF V600E + MLH1 loss largely excludes Lynch — but **~1% of Lynch is BRAF V600E**

STAGING — AJCC 8TH

T1 / **T2** / **T3**	submucosa / through muscularis propria / into pericolorectal tissue
T4a / **T4b**	through visceral peritoneum / invades adjacent organ
N1 / **N2**	1–3 nodes (N1c = deposits only) / **≥4 nodes**
M1a / **b** / **c**	1 site / ≥2 sites / **peritoneal**
Stage III	any node-positive, M0

KEY TRIALS

IDEA	3 vs 6 mo adjuvant — **failed non-inferiority overall** (HR 1.07, margin 1.12), but **3 mo CAPOX non-inferior while FOLFOX favored 6 mo** (Grothey, NEJM 2018)
ATOMIC	adding IO to adjuvant chemo in dMMR (ASCO 2025)
KEYNOTE-177	1L pembrolizumab in dMMR: PFS **16.5 vs 8.2 mo**
BREAKWATER	1L BRAF V600E: FOLFOX + encorafenib + cetuximab
BEACON	BRAF V600E later line (Kopetz, NEJM 2019)

TREATMENT PATHWAY

Localized — stage I to III

Surgery first
colon is resected upfront (unlike rectal)
pathology drives the adjuvant decision
no adjuvant therapy for stage I

Stage II (node-negative)
risk factors: T4, LVI, PNI, obstruction, perforation, poor differentiation, <12 nodes
low risk observe; RFs 5-FU ± oxaliplatin

Stage III
virtually all receive adjuvant chemotherapy
age <75: fluoropyrimidine + oxaliplatin
age >75: fluoropyrimidine alone
IDEA: 3 mo CAPOX ok — FOLFOX needs 6 mo
high risk (T4 and/or N2) favors 6 months

dMMR / MSI-H

Immunotherapy-sensitive
consider neoadjuvant IO before resection
adjuvant: add IO to chemo (ATOMIC)
adjuvant 5-FU alone may be HARMFUL

Metastatic

Oligometastatic
liver/lung only: surgical consult upfront
FOLFOX or CAPEOX preferred — cure possible

1L by molecular subtype
dMMR: pembrolizumab (KEYNOTE-177)
BRAF V600E: FOLFOX + encorafenib + cetuximab
left-sided RAS/BRAF WT: doublet + anti-EGFR
right-sided or RAS-mut: FOLFOXIRI ± bev

Later lines
stop oxaliplatin at ~4 months (neuropathy)
swap backbone; HER2 matters in 2nd line

> **Oxaliplatin neuropathy is the dominant long-term toxicity.** Six months may raise the risk of permanent neuropathy as much as **tenfold** vs three months. Do not exceed ~4 months in initial metastatic therapy (OPTIMOX); never given as a single agent.

REGIMENS & DOSING

Regimen	Dosing	Cycle
FOLFOX	oxaliplatin 85 d1 · 5-FU 1200–1500/day CIV d1–2	q14d
CAPEOX / **XELOX**	oxaliplatin 130 d1 · capecitabine 1000 BID d1–14	q21d
FOLFIRI	irinotecan 180 d1 · 5-FU 1200–1400/day CIV d1–2	q14d
FOLFOXIRI	oxaliplatin 85 · irinotecan 165 · 5-FU 1200–1400/day CIV d1–2	q14d
Bevacizumab	7.5 mg/kg (q3wk) or 5 mg/kg (q2wk)	q21d / q14d
Cetuximab / **panitumumab**	cetuximab 400 load then 250 weekly (or 500 q2wk) · panitumumab 6 mg/kg	wkly / q2wk
Regorafenib	ramp 80→120→160 mg weekly in C1; then 160 mg d1–21	q28d
Trifluridine/tipiracil	35 mg/m² PO BID d1–5 and d8–12	q28d
Fruquintinib	5 mg PO daily × 21 days	q28d

All doses mg/m² unless stated. **Institutional preference: no 5-FU bolus** (higher toxicity) — which also removes the need for leucovorin as a stabilizer.

BIOMARKER → THERAPY

Alteration	Therapy
dMMR / **MSI-H**	**Pembrolizumab** 200 mg q3wk or 400 mg q6wk — 1L metastatic; also neoadjuvant/adjuvant
BRAF V600E, 1L	FOLFOX + **encorafenib** 300 mg daily + **cetuximab**
BRAF V600E, later	Encorafenib + cetuximab; triplet adds binimetinib
RAS WT + left-sided	Cetuximab or panitumumab — **no benefit if right-sided or RAS-mutant**
HER2 amplified	Trastuzumab + pertuzumab; **tucatinib** 300 BID + trastuzumab; **T-DXd**
NTRK fusion	Larotrectinib 100 mg BID · entrectinib 600 mg daily

SURVEILLANCE

- H&P + **CEA** q3–6 mo × 2 yr, then q6 mo — 5 years total
- CT C/A/P: stage II–III q6–12 mo × 5 yr; stage IV q3–6 mo × 2 yr then q6–12 mo
- Colonoscopy at **1 year** post-op; within 3–6 mo if no complete pre-op scope
- If adenoma found repeat in 1 yr; otherwise 3 yr, then every 5 yr

PEARLS

- **Irinotecan** with great caution in the elderly, biliary obstruction, or hyperbilirubinemia
- **Regorafenib** is very toxic — many do better at lower doses; **bevacizumab** benefit is modest
- **ctDNA timing conflict:** results take 4–6 wk but adjuvant works best within 6 wk of surgery
- Metastatic **rectal** regimens are identical to metastatic CRC